

The Hierarchy ν/E_p

Two Derivations Compared: FFGFT and Observer Patch Holography

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22 September 2026

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The Hierarchy ν/E_P : Two Derivations Compared

Abstract. FFGFT derives the electroweak scale from ξ (R104) and reaches the Planck scale through its own chain $\xi \rightarrow G \rightarrow E_P$. FFGFT therefore delivers the hierarchy number ν/E_P entirely by itself; the electron mass serves only as the SI anchor. Observer Patch Holography (OPH) delivers the same number conditionally through its Theorem A. The two derivations give 2.0389×10^{-17} (FFGFT, +1.10 %) and 2.0200×10^{-17} (OPH, +0.16 %). The FFGFT deviation is entirely the known bare gap of the lepton ladder (Doc. 352); the Planck chain contributes less than 10^{-4} . The FFGFT entry for a hashed comparison table is given.

List of symbols

[B]	algebraically proven
[K]	numerically confirmed
[S]	structurally plausible, derivation open
[Q]	taken from an external source, cited

Symbol	Meaning
ξ	FFGFT scaling parameter (4/30000)
ν	electroweak scale
E_P	Planck energy (not reduced)
$G, C_{\text{conv}}, K_{\text{frak}}$	gravitational constant and factors of the SI representation (Docs. 012/013)
E_*, P, α_U	OPH reference energy, pixel ratio, unified coupling

1 Occasion

Doc. 374 brought FFGFT and OPH together at the electroweak scale and, in doing so, combined m_e/ν from the FFGFT ladder with ν/E_* from OPH. That understates the FFGFT side: FFGFT derives ν itself (R104) and reaches E_P through its own chain (Doc. 374, § 5). The hierarchy number ν/E_P is therefore not a combination but a second, independent derivation of the same quantity. This document places the two derivations side by side.

2 The FFGFT derivation

1. Electroweak scale (R104, Doc. 006):

$$\nu = \frac{m_e}{\frac{4}{3} \xi^{3/2}} = 248.93 \text{ GeV}. \quad (1)$$

2. Planck energy through its own chain (Docs. 012, 013, 180): $G = \xi^2/(4m_e) \cdot C_{\text{conv}} \cdot K_{\text{frak}}$, hence $E_P = 1.22087 \times 10^{19}$ GeV.
3. Hierarchy:

$$\left. \frac{v}{E_P} \right|_{\text{FFGFT}} = 2.0389 \times 10^{-17} \quad \textbf{[K]}. \quad (2)$$

Against the SM value $v = 246.22$ GeV (definition from G_F , R112) this is +1.10 % higher. The deviation is entirely that of v ; the Planck chain contributes less than 10^{-4} . It is identical to the bare gap of the lepton ladder m_e/v (−1.09 %, Doc. 352): v is obtained from the bare relation and inherits its residual. Not a new finding.

(Added on 29 Sept. 2026: this route uses two measured values, m_e for v and, via C_{conv} , the measured value of G for E_P . Directly and without an anchor, v/E_P follows from Doc. 149: $v/E_P = \xi^4/(5\pi) = 2.0120 \times 10^{-17}$, −0.23 %. The factor 1.0134 between the two values is the residue of the electron mass in the chain. See Doc. 383.)

3 The OPH derivation [Q]

Theorem A from code/particles/hierarchy/theorem_package.md:

$$\frac{v}{E_\star} = P^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U(P)}\right] = 2.0200 \times 10^{-17}, \quad (3)$$

reproduced in the check script. The derivation is explicitly conditional in OPH; the identification $E_\star = E_P$ is a reading **[S]** (Doc. 374). Against the SM value: +0.16 %.

4 Comparison

Derivation	v/E_P	vs. SM	Origin of the deviation
FFGFT	2.0389×10^{-17}	+1.10 %	bare residual of the ladder (Doc. 352)
OPH [Q]	2.0200×10^{-17}	+0.16 %	conditional fixed point, reading $E_\star = E_P$
SM comparator	2.0167×10^{-17}	—	v from G_F (R112)

The two derivations lie 0.93 % apart. They also differ in sensitivity:

- FFGFT: $\partial \ln v / \partial \ln \xi = -3/2$ **[B]**. A rational structural quantity with power 3/2; small changes of ξ act weakly.
- OPH: $\partial \ln(v/E_\star) / \partial \alpha_U = \pi/(2\alpha_U^2) \approx 929$. The coupling sits in the exponent; the 0.16 % corresponds to a shift of α_U by 4×10^{-5} relatively.

The FFGFT value is robust and carries a known residual; the OPH value is more accurate but highly sensitive and conditional. Both statements are testable: on the FFGFT side through the resolution of the bare residual (Doc. 352), on the OPH side through the clock attachment and the identification of $E_\star$.

5 Entry for the comparison table

For a hashed comparison table the FFGFT entry, including the route, reads:

```
v/E_P | FFGFT | 2.03894e-17 | route: v=m_e/((4/3)xi^(3/2)) (R104, Doc 006) ;
E_P via G=xi^2/(4m_e)C_conv K_frak (Docs 012/013/180) ; bare ladder
```

SHA-256 of the string (generated in the check script):

```
84fac85fc55d8c5928a07550e774683961613059415b0ef02715b327f58fb15f
```

Tool and method

Claude (Anthropic) was used as a tool for: writing and running the Python check script, drafting the $\text{\LaTeX}$ sources, and maintaining the changelog. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. OPH values are taken verbatim from the public repository; measured values (CODATA 2022, PDG) serve only as comparators.

6 Result

1. FFGFT delivers $\nu/E_P = 2.0389 \times 10^{-17}$ entirely by itself: ν from ξ (R104), E_P from its own chain **[K]**.
2. The deviation of $+1.10\%$ is the bare residual of the lepton ladder; the Planck chain contributes less than 10^{-4} **[K]**.
3. OPH delivers, conditionally, 2.0200×10^{-17} ($+0.16\%$) **[Q]**; the derivations lie 0.93% apart.
4. Sensitivity: FFGFT $-3/2$ in $\ln \xi$ **[B]**, OPH ≈ 929 in α_U .
5. FFGFT table entry given with route and SHA-256.

Check script: 2/python/Dok375_Skripte/pruef_375_hierarchie.py (11/11)

Register (Doc. 190): no entry.

Bibliography

- [1] J. Pascher, *Doc. 006 — T_0 particle masses* (2025).
- [2] J. Pascher, *Doc. 012 — T_0 gravitational constant* (2025).
- [3] J. Pascher, *Doc. 013 — T_0 and SI units* (2025).
- [4] J. Pascher, *Doc. 180 — Derivation of L_0* (2026).
- [5] J. Pascher, *Doc. 352 — Lepton residuals* (2026).
- [6] J. Pascher, *Doc. 374 — Common anchor* (2026).
- [7] B. Mueller et al., *Observer Patch Holography*, github.com/FloatingPragma/observer-patch-holography (as of 22 September 2026).